

Capturing the Supply Chain as HBOMs, SBOMs and more with SPDX 3.1

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Take Away Message

Supply Chains need Universal Data Capture and Sharing

System Package Data **Exchange** (SPDX)

SPDX is a graph language with an integrated ontology for universal data capture and sharing.

Static Data
Management

Dynamic Data
Management

Universal Data
Merging



Operational Solutions

Topics

SPDX 3.1 Overview

- SPDX history
- New profiles
- Action ontology: Defined process, actions and requirements
- SPDX ontology used as part of the Graph Language

Supply Chain Integration SPDX 3.1

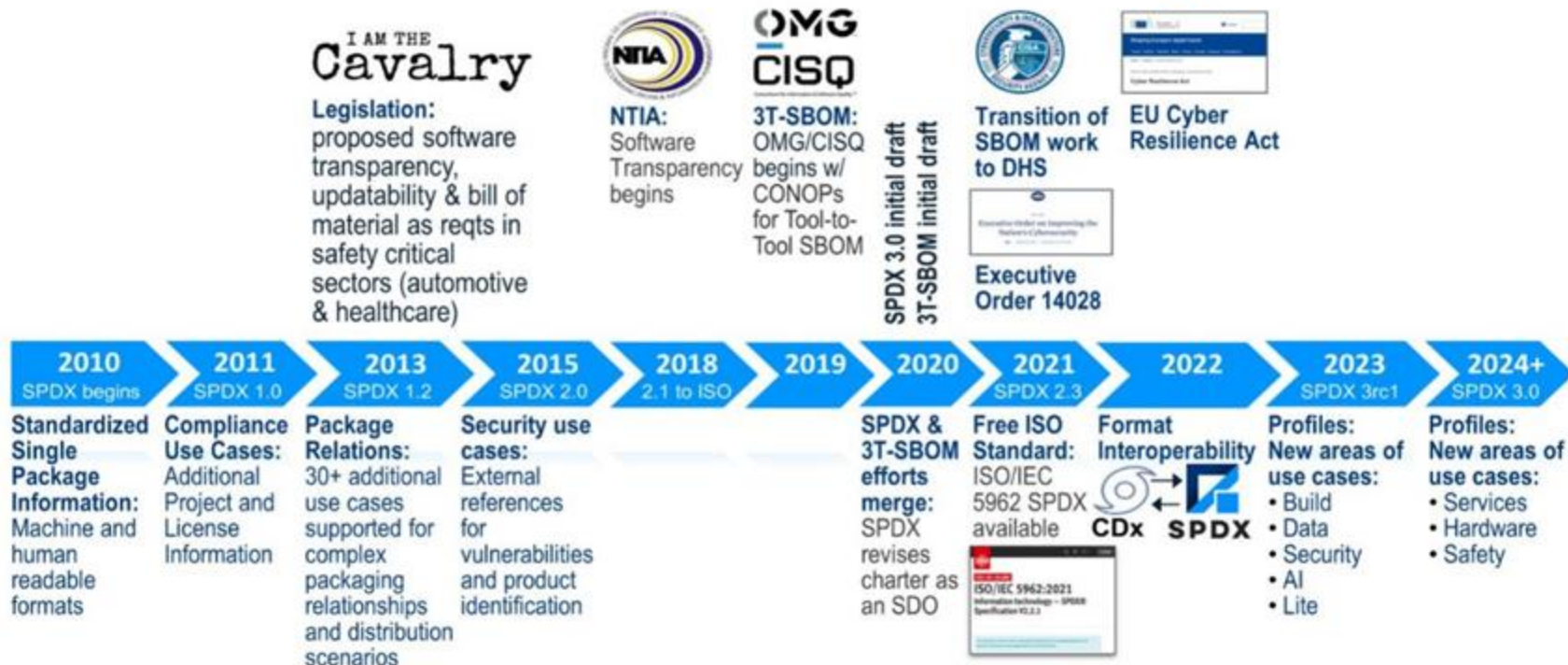
- Supply Chain methodology
- Supply Chain traceability
- Reading Supply Chain examples
- Regulatory compliance with SPDX



SPDX History



Timeline of SPDX Evolution - Use Case by Use Case



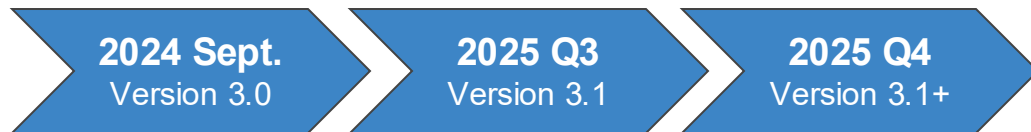
MITRE



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SPDX 3.1 Profiles

Additional Timeline Items



Engineering Ontology
Graph Language
Cross Profile Integration
Operational Solutions

Profiles: New

- Safety
- Operations
- Dataset
- AI
- Hardware
- Supply Chain

Profiles: 3.1+ Additions

- Cryptology
- Security
- Threat and Controls



SPDX 3.1+ Profiles

SPDX is a graph language driven by relationships

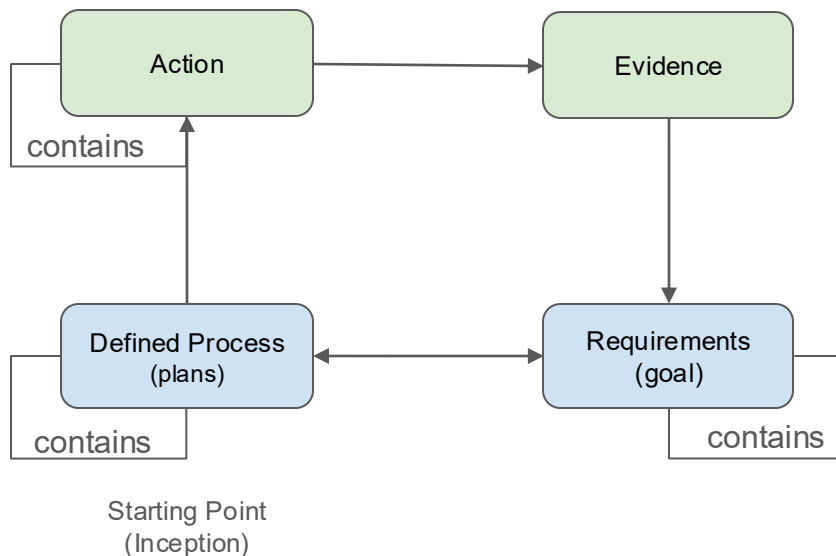
- Core
- Software
- **Hardware**
- **Supply Chain**
- Licensing
- Services
- Operations
- AI
- Dataset
- Security
- Cryptology
- Safety
- Threat Analysis
- Lite

SPDX as a Knowledge Graph

Empowering Data Exchange



SPDX 3.1 is built upon abstraction for optimal information capture and relationship definition.



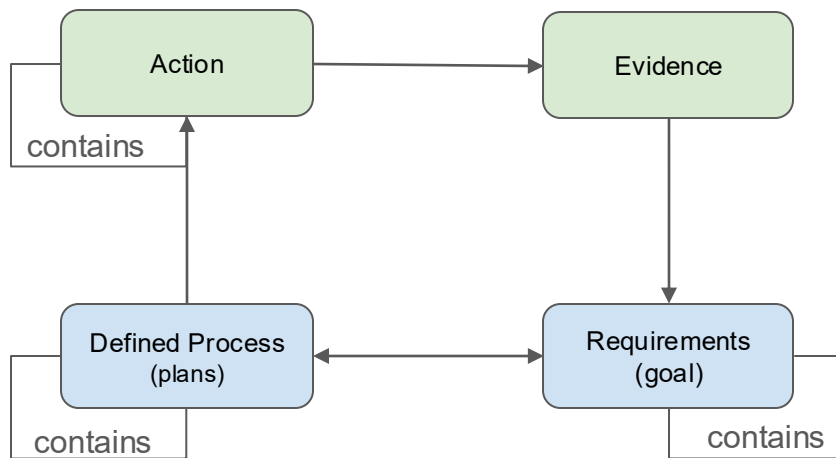
Design Assurance abstraction

SPDX as a Knowledge Graph

Addition of Action Methodology



The **requirements** to define a product. **Processes** define how. **Actions** is the execution of processes. **Data** is who, how, what and when which is evidence. **Evidence** allows you make decisions.

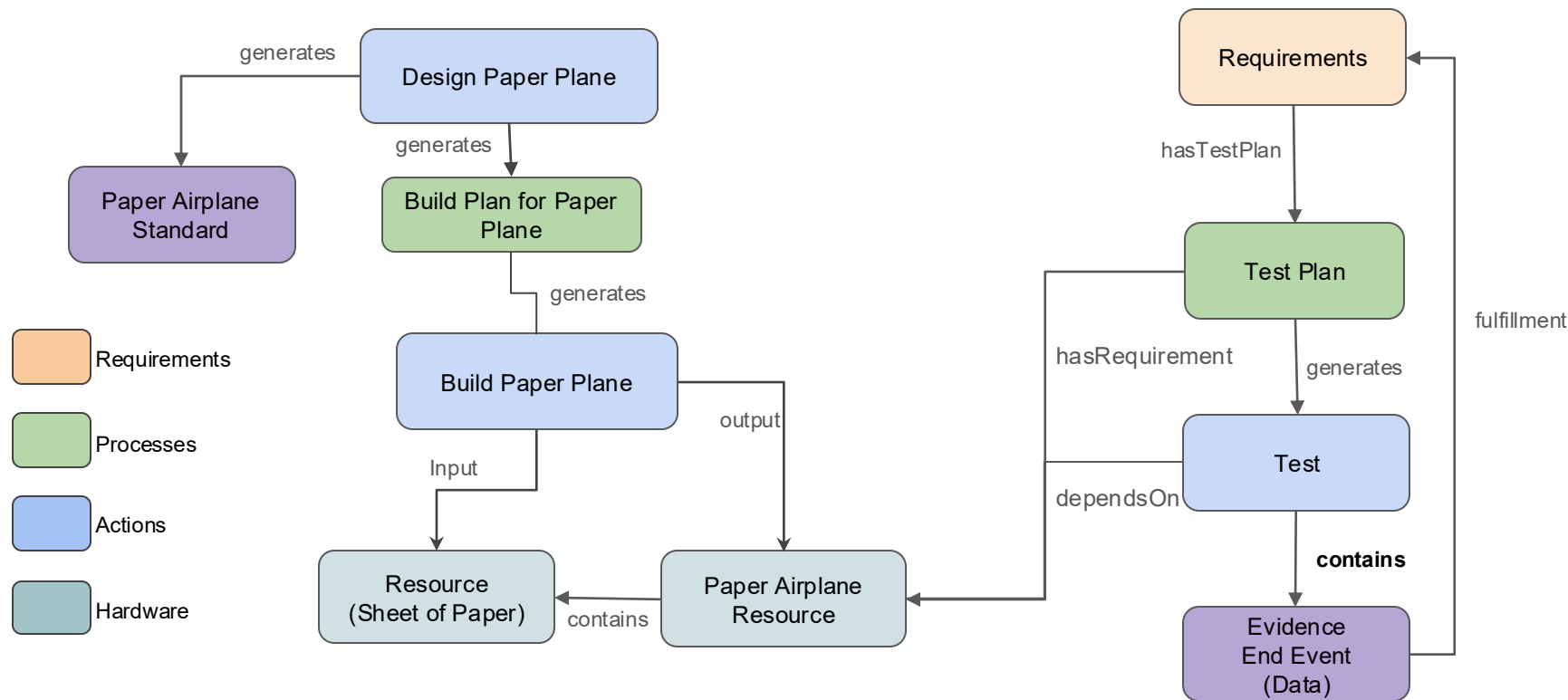


Starting Point
(Inception)

Modelling the building of a paper airplane



Process (plan) of building is followed by Action (doing).





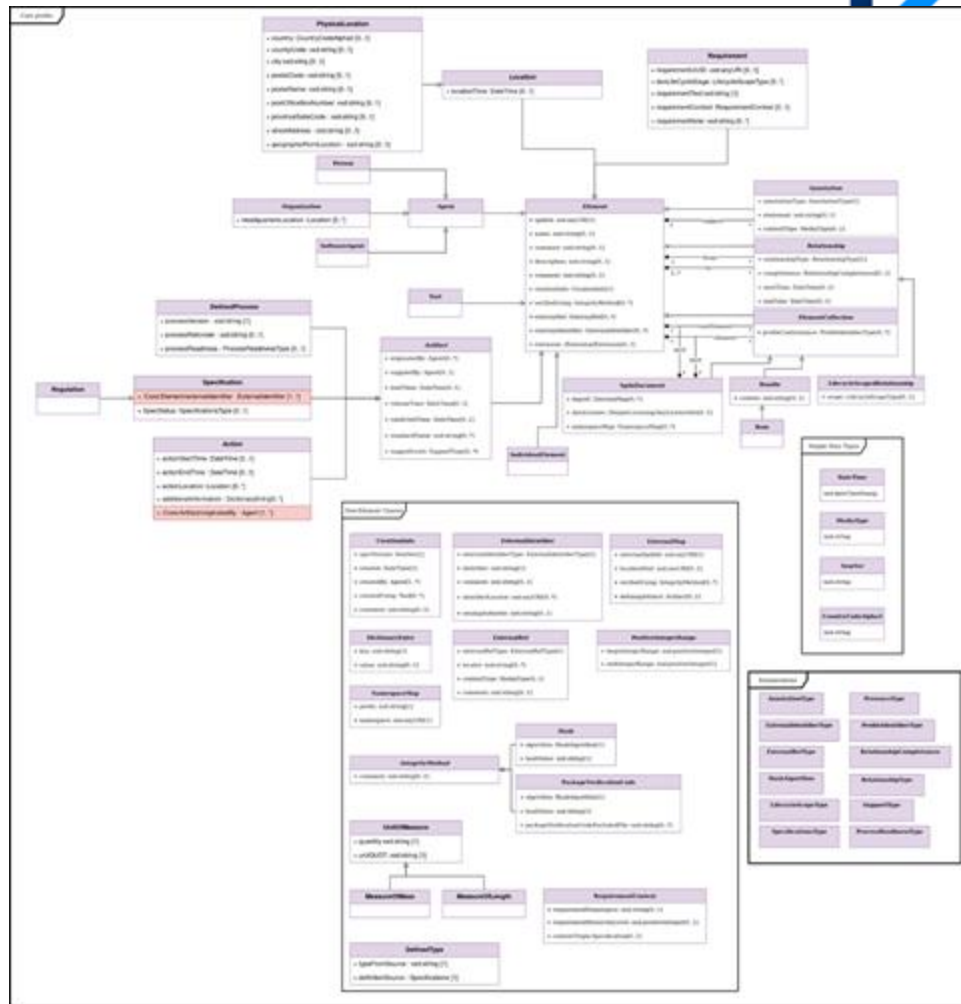
SPDX 3.1 Core

Core Elements

- **Requirements**
- Relationships
- Agents
- Location
- **Defined Processes**
- **Specification**
- **Action**
- Organization
- Record Creation Information
- Integrity Method

Core is used by all profiles.

<https://github.com/spdx/spdx-3-model/blob/develop/images/model-Core.png>



Hardware Profile



Hardware

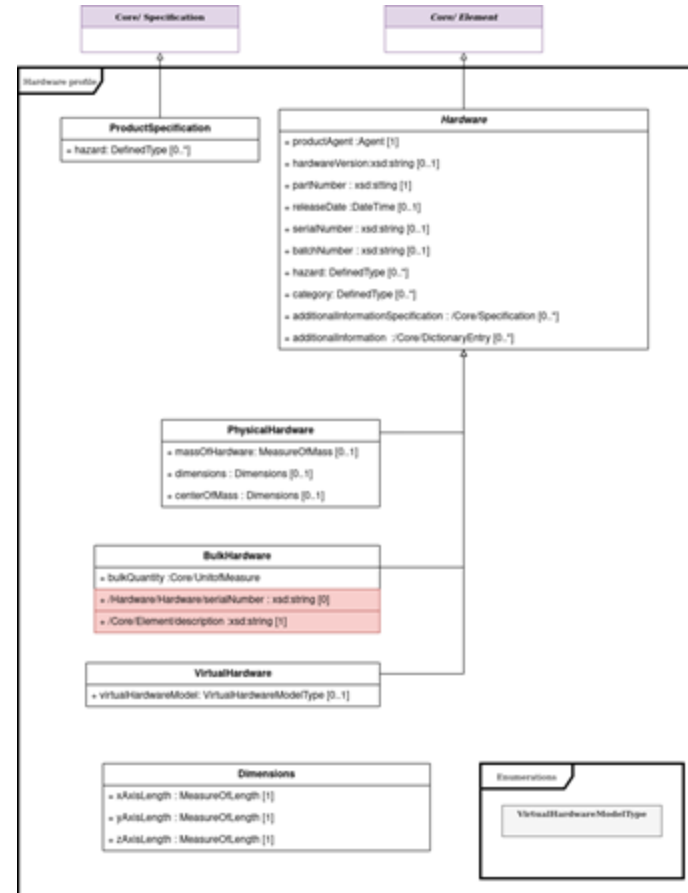
- Physical
- **Virtual**
- Bulk

Hardware Description

The Hardware namespace defines metadata related to physical and virtual hardware properties.

Hardware is any product, real or virtual. A product is tangible and is the result of labor, or of a natural or artificial process.

<https://github.com/spdx/spdx-3-model/blob/develop/images/model-Hardware.png>



Supply Chain Profile

Creation

- Manufacture
- Assemble
- Reproduce
- Harvest

Modify

- Change
- Boundary

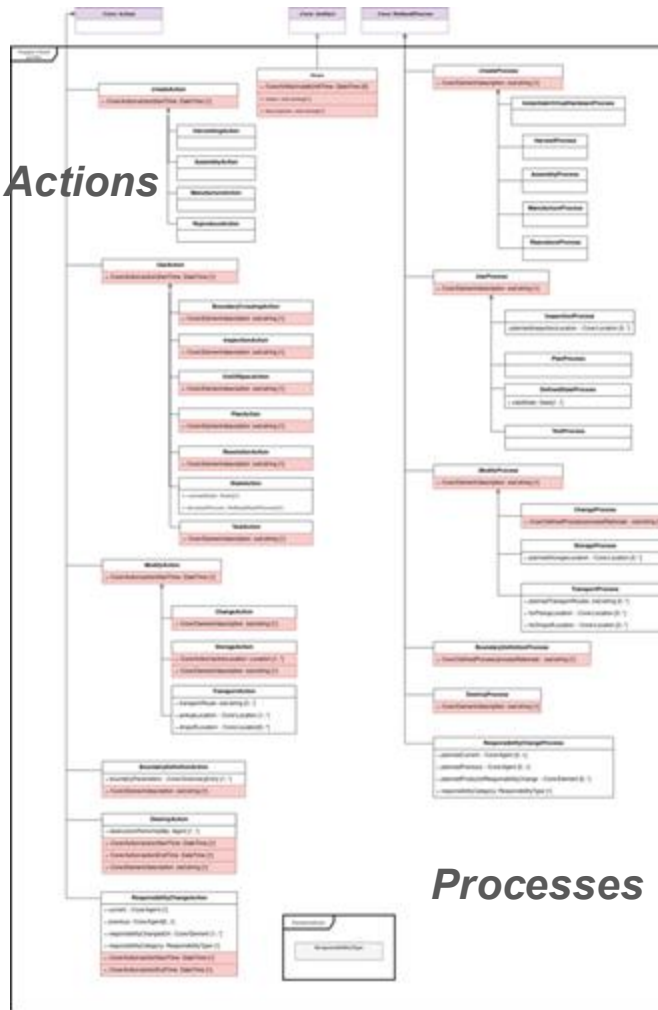
Destroy

The Supply Chain profile documents events and processes associated with the lifecycle of a product, including its creation, transportation, usage, and decommissioning.

<https://github.com/spdx/spdx-3-model/blob/develop/images/model-SupplyChain.png>



Actions



Processes

**SPDX
HARDWARE**

Responsibility

- Owner
- Custody

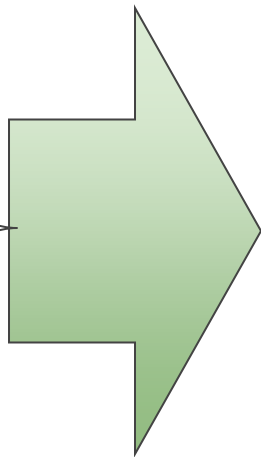
Use:

- Inspection
- Plan
- Resolution
- Boundary
- Out of Spec

SPDX Profile Integration

Domain Profile Knowledge Sharing

- Core
- Software
- Hardware
- Supply Chain
- Licensing
- Services
- Operations
- AI
- Dataset
- Security
- Cryptology
- Safety
- Threat Analysis



One Document with multiple profiles

```
{
  "@context": "https://spdx.github.io/spdx-spec/v3.0.1/rdf/spdx-context.jsonld",
  "@graph": [
    {
      "spdxId": "urn:nano id:BNE ris92wBqwo72YI5C04",
      "type": "hardware_PhysicalHardware",
      "creationInfo": "._:b0",
      "name": "AMD A10-6700 APU with Radeon(tm) HD
Graphics ",
      "hardware_partNumber": " A10-6700 APU",
      "hardware_productAgent":
"urn:nano id:YMYzgKjNNuJ20iQN-NuL9"
    },
    {
      "type": "CreationInfo",
      "created": "2025-07-22T03:32:00Z",
      "createdBy": "urn:nano id:Dp9d5Qlhf6JDBHeNBvW-7",
      "specVersion": "3.0.1",
      "@id": "._:b0"
    },
    {
      "spdxId": "urn:nano id:YMYzgKjNNuJ20iQN-NuL9",
      "type": "Organization",
      "creationInfo": "._:b0",
      "name": "AMD"
    },
    {
      "spdxId": "urn:nano id:4yT0oV6gTaRVweNrYle cQ",
      "type": "Relationship",
      "creationInfo": "._:b0",
      "from": "urn:nano id:Om vGSTuydsi7YK1qT5Rgg",
      "relationshipType": "hasOutput"
      "to": "urn:nano id:BNE ris92wBqwo72YI5C04"
    }
  ]
}
```

SPDX makes it easy to read and link

Use Case: Extracting Information



SPDX Profile

Hardware

Hardware



Build, AI, Software

Software



Connections

Network

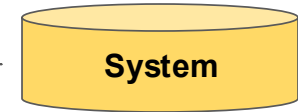


**Operations,
Software Agent**

Processes



Define systems



Risk mitigation Analysis
(Tools such as NSCU)

Linking Hardware Information & SC

Gathering and enhancing xBOM information with SC information



Category

System Auditing

SPDX Core Profiles

SPDX Supply Chain

Tools

Collection
Presentation
Amalgamation
Analysis
Operations

Core
SBOM
HBOM
AIBOM
Operations
Licensing
Security

Origin
Traceability
Buy/Sell chain
Compliance

Focus

Hardware Owner

Inventory
Operations
Maintenance
Risk Mitigation

Information Platform

Data Sharing
Standard
Language Graph
Risk Mitigation

Information Producer

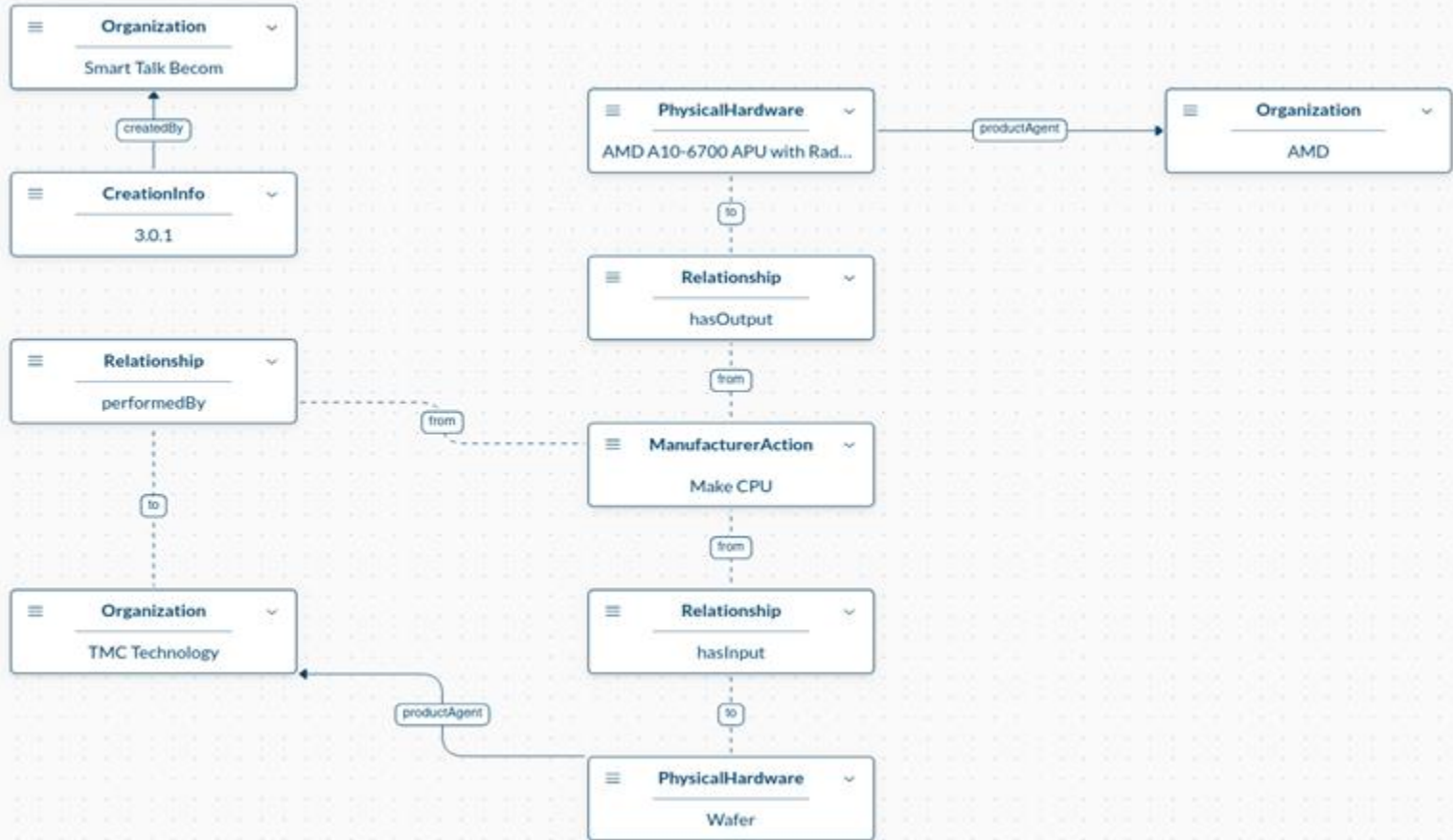
Link information
from multiple
sources for use
by chain



Information Flow

SPDX makes it easy to read and link

Use Case: Viewing Information



SPDX Supply Chain Solution -

Open Source Solution with SPDX

SPDX solution is an **open-standard, ontology-based** platform and **framework** enabling seamless, secure data exchange and regulatory compliance.

Trusted by leaders like Microsoft and AMD.

Supply Chain Need Categories

Supply Chain Regulatory Requirements by Category



- Sustainability/Human Rights: EU CSDDD, Germany LkSG, UK Modern Slavery Act
- Trade & Customs: WCO SAFE, CTPAT, UN/CEFACT
- Resilience/Continuity: ISO 22301, ISO 28000, EO 14017
- Traceability & Data: GS1, ISO 307
- Cybersecurity: NIST 800-161, ISO/IEC 27036, EO 14028, EU CRA, CMMC

Supply Chain Definition

Supply Chain Challenge

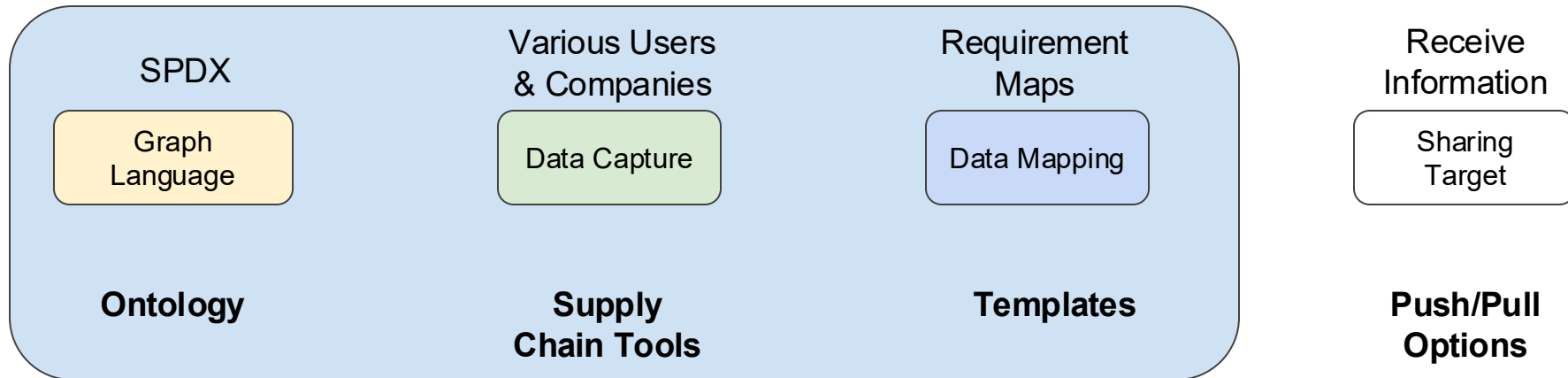
A supply chain is the entire network of organizations, people, activities, information, and resources involved in moving a product or service from supplier to customer. It encompasses **every phase**, from sourcing **raw materials and components**, through **manufacturing and assembly**, to **warehousing, distribution, and final delivery** to the end user.

TODAY: Is it possible for each organization to capture their part of the supply chain, securely *share the information with multiple stakeholders up and down the chain* in a **simple and cost effective** within a changing regulatory compliance structure?

YES with SPDX!

Supply Chain Core Elements

Data Process Stages



Supply Chain And SPDX

Example Supply Chain And Visualization in SPDX



Let look a Graphite mining org->(SPDX Organization) producing Graphite -> (Mining is a

Organization

Summary

A group of people who work together in an organized way for a shared purpose.

Description

An Organization is a group of people who work together in an organized way for a shared purpose.

Metadata

<https://spdx.org/rdf/3.0.1/terms/Core/Organization>

Instantiability	Concrete
SubclassOf	Agent

HarvestAction

Summary

HarvestAction represents the act of creating a product by directly extracting goods or materials from nature.

Description

HarvestAction refers to an event where goods or products are obtained directly from natural sources. This includes activities such as mining, and fishing. In such cases, there may be no associated hasInput relationships, as the products are not assembled from other components but sourced directly.

Metadata

<https://spdx.org/rdf/3.1/terms/SupplyChain/HarvestAction>

Instantiability	Concrete
SubclassOf	CreateAction

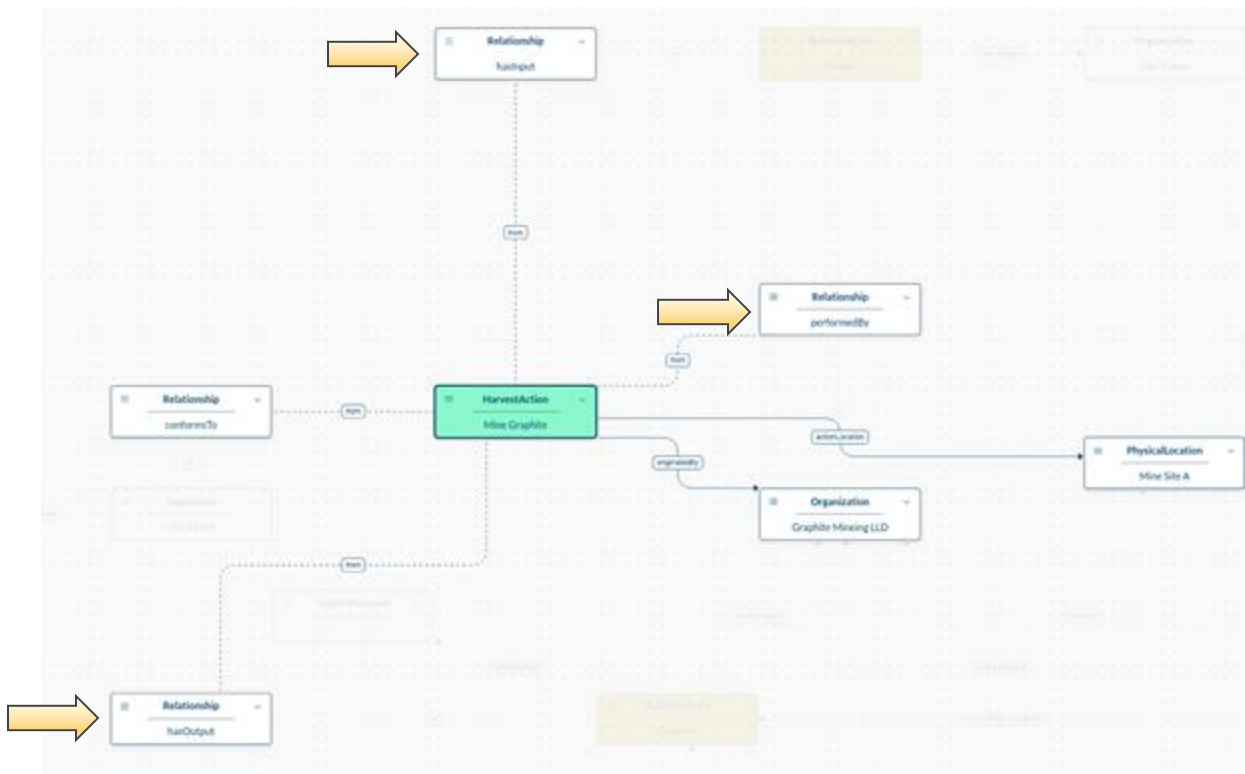
Example Supply Chain



Supply Chain And SPDX

Example Supply Chain

This HarvestAction can have relationships to other SPDX Items.



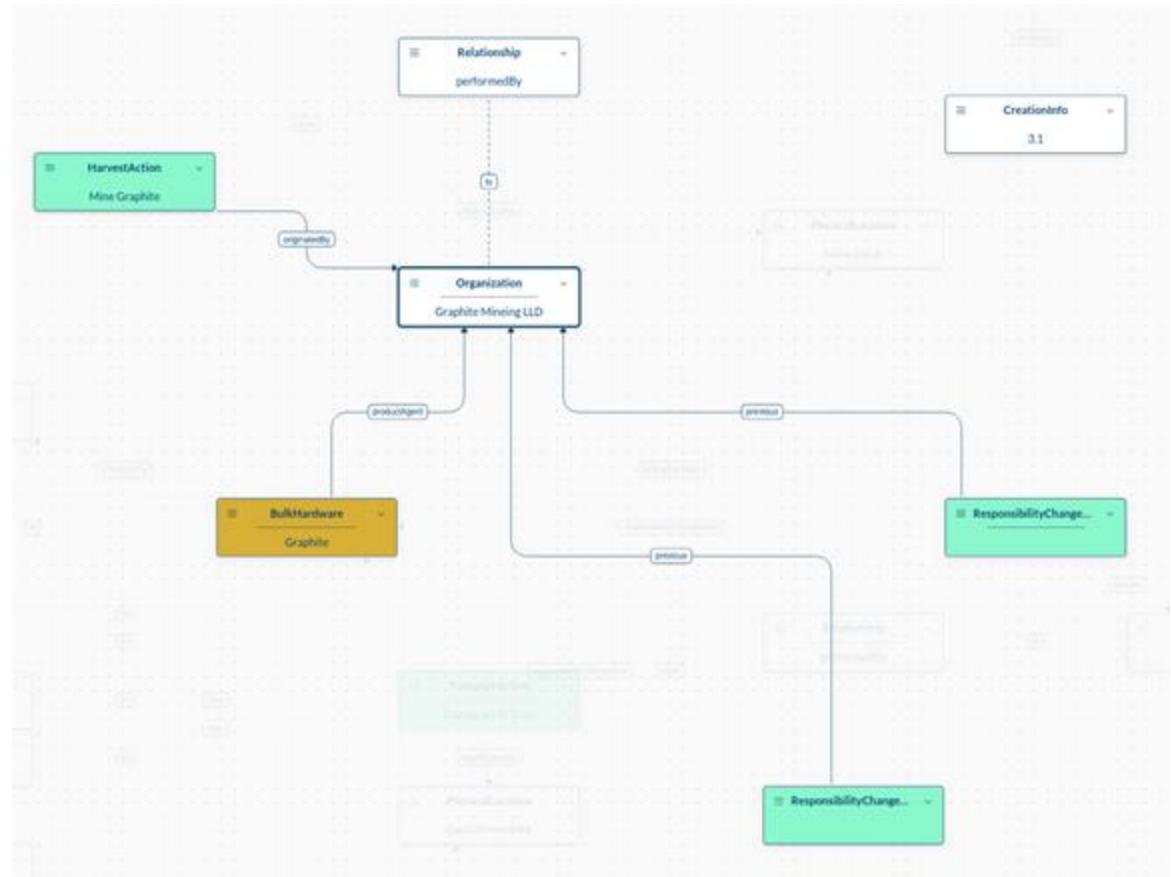
Supply Chain And SPDX

Example Supply Chain

Organization

Organization relationships define the role an organization plays in relation to any other elements in the graph.

This is flexible and powerful.



Supply Chain And SPDX

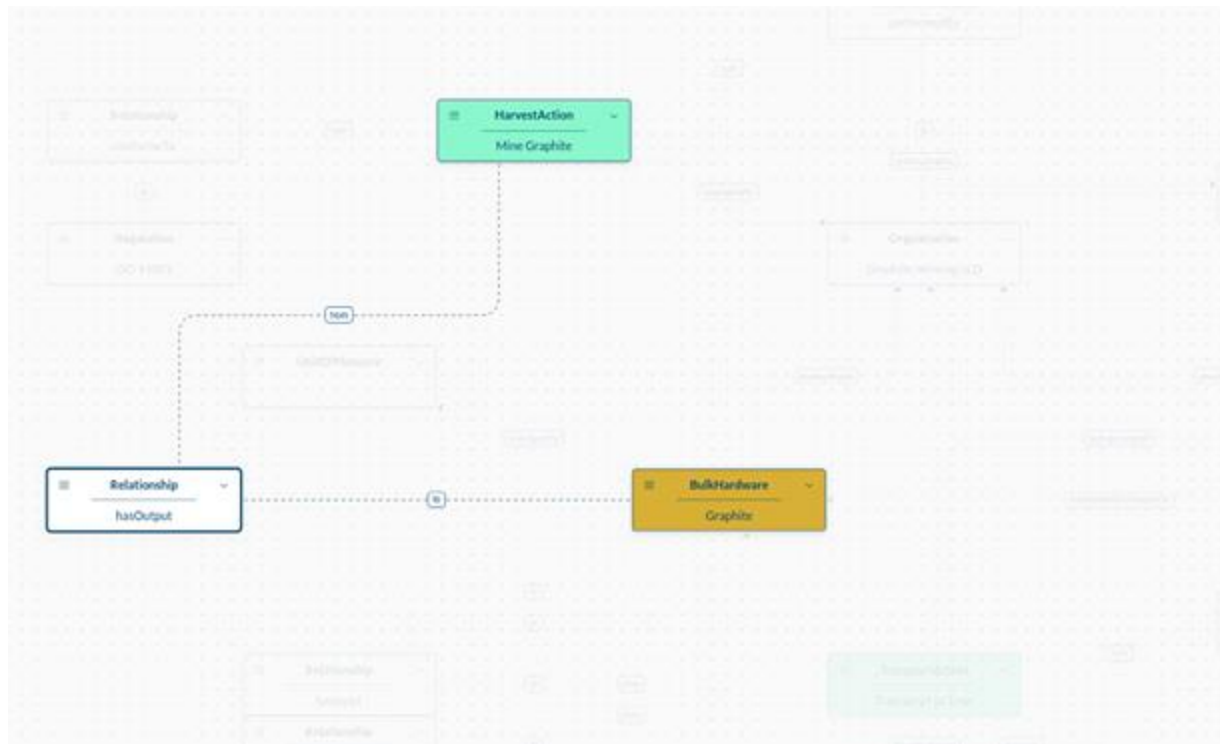
Example Supply Chain



Consider the Output relationship.

Harvest Action produces BulkHardware.

Bulk hardware (product) is identified in bulk units as opposed to individual units.



Supply Chain And SPDX

Example Supply Chain



BulkHardware
measurement units can
be used to record CO²
emissions per action.

QUDT: <https://qudt.org/>

BulkHardware

Summary

Products or commodities produced as a bulk unit are called bulk products. Commodities are often sold in bulk.

Description

Products or commodities produced as a bulk unit are called bulk products. Commodities are often sold in bulk based on a batch or bulk ID number associated with the bulk unit. Bulk units defined in the QUDT Units standards.

Metadata

<https://spdx.org/rdf/3.1/terms/Hardware/BulkHardware>

Instantiability	Concrete
SubclassOf	Hardware

Supply Chain And SPDX

Example Supply Chain



Bulk Hardware units and measurement are defined in bulk Quantity.

/Core/UnitOfMeasure

bulkQuantity

Summary

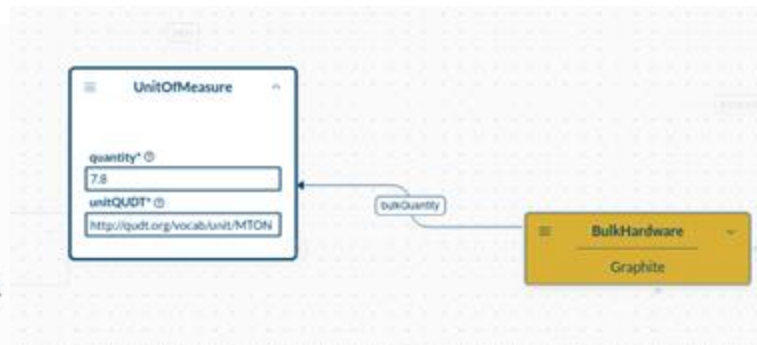
The amount or measure of a bulk product.

Description

The amount or measure of a bulk product includes the quality and unit of measure.

Metadata

<https://spdx.org/rdf/3.1/terms/Hardware/bulkQuantity>



Supply Chain And SPDX

Example Supply Chain



Representing a

TransportAction

Summary

An actual change to a product's location.

Description

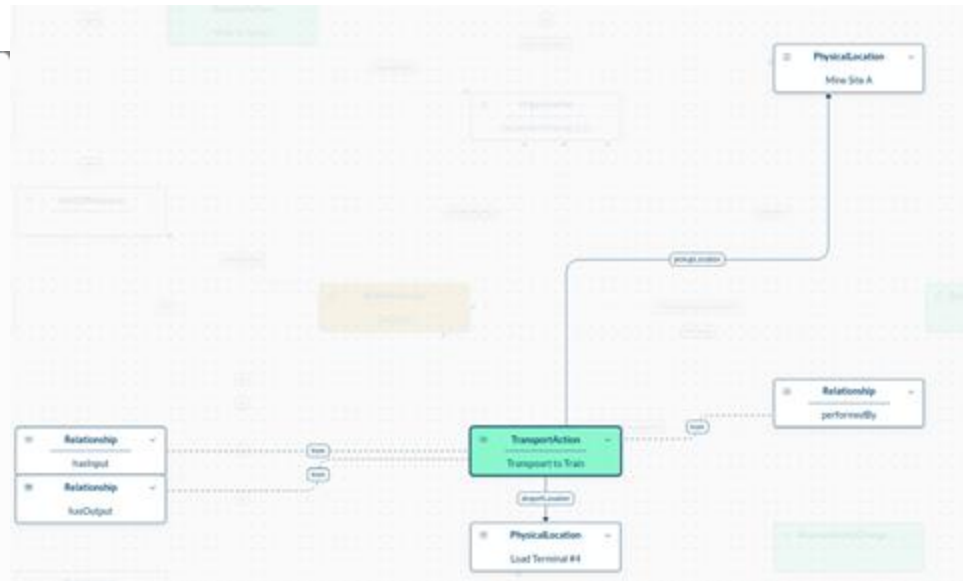
A TransportAction expresses the change in location of a product, such as a component being moved from an assembly plant to a warehouse.

Metadata

<https://spdx.org/rdf/3.1/terms/SupplyChain/TransportAction>

Instantiability Concrete

SubclassOf [ModifyAction](#)



Supply Chain And SPDX

Example Supply Chain



Representing a
TransportAction

➡ Dropoff or pickup links

dropoffLocation

Summary

The location for dropping off or delivering a package or item.

Description

This is a specific location that an item was dropped off to.

Metadata

<https://spdx.org/rdf/3.1/terms/SupplyChain/dropoffLocation>

pickupLocation

Summary

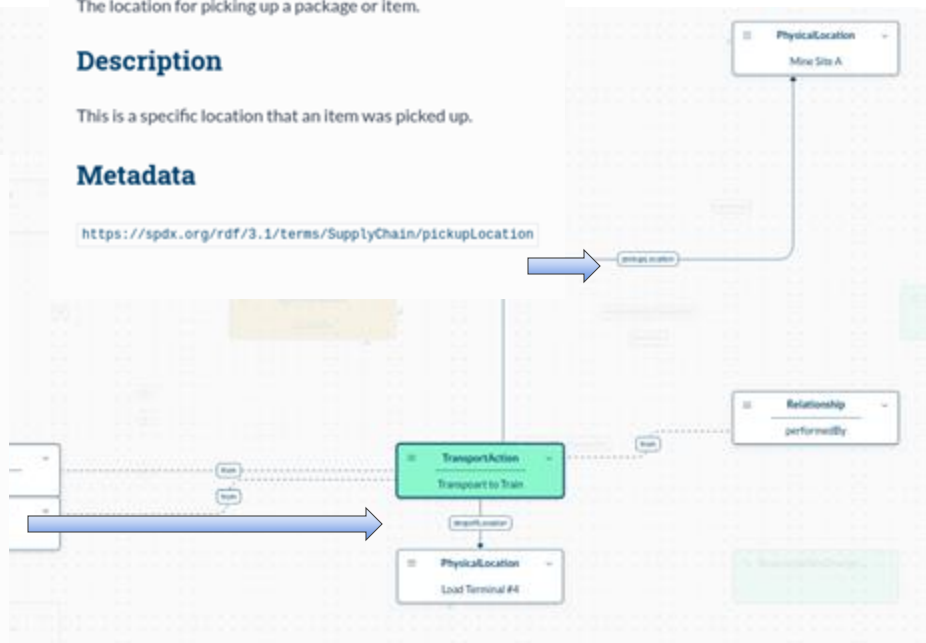
The location for picking up a package or item.

Description

This is a specific location that an item was picked up.

Metadata

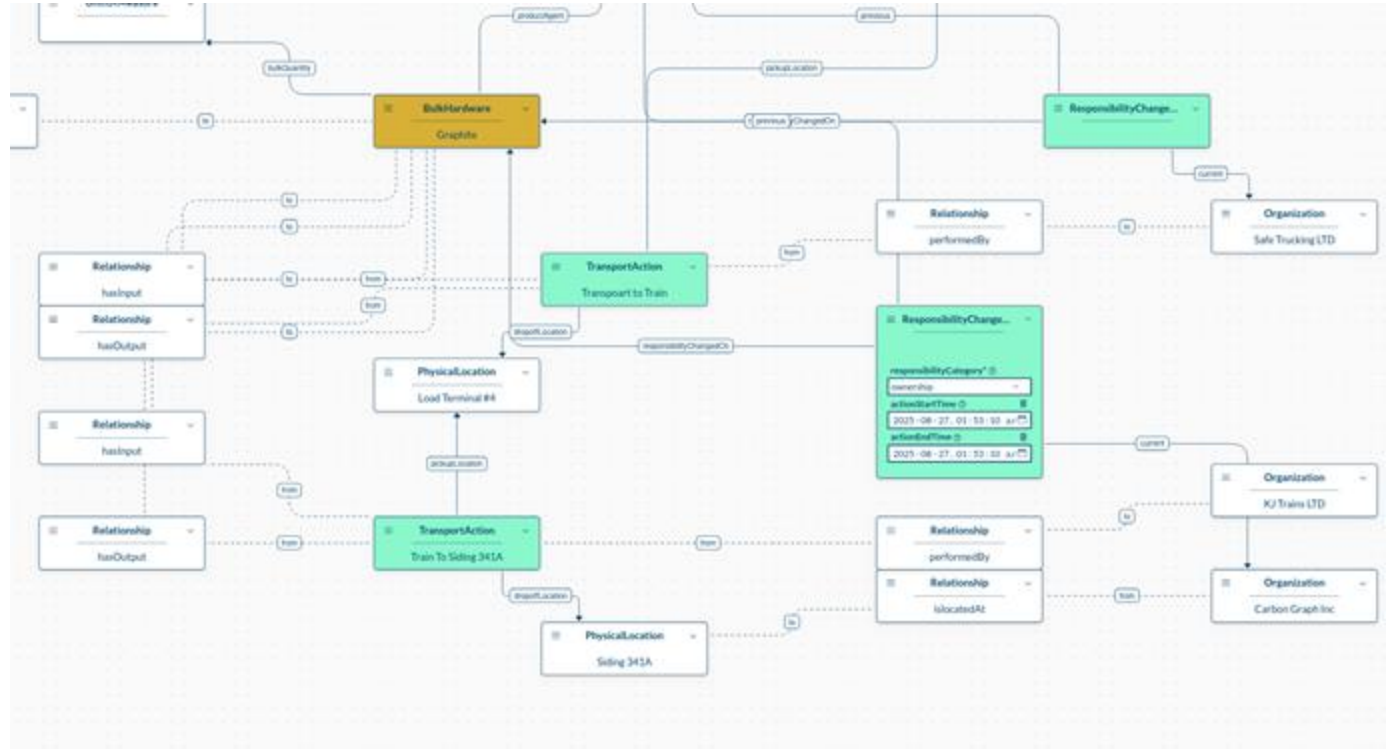
<https://spdx.org/rdf/3.1/terms/SupplyChain/pickupLocation>



Supply Chain And SPDX

Example Supply Chain

TransportAction can
link to new
TransportAction

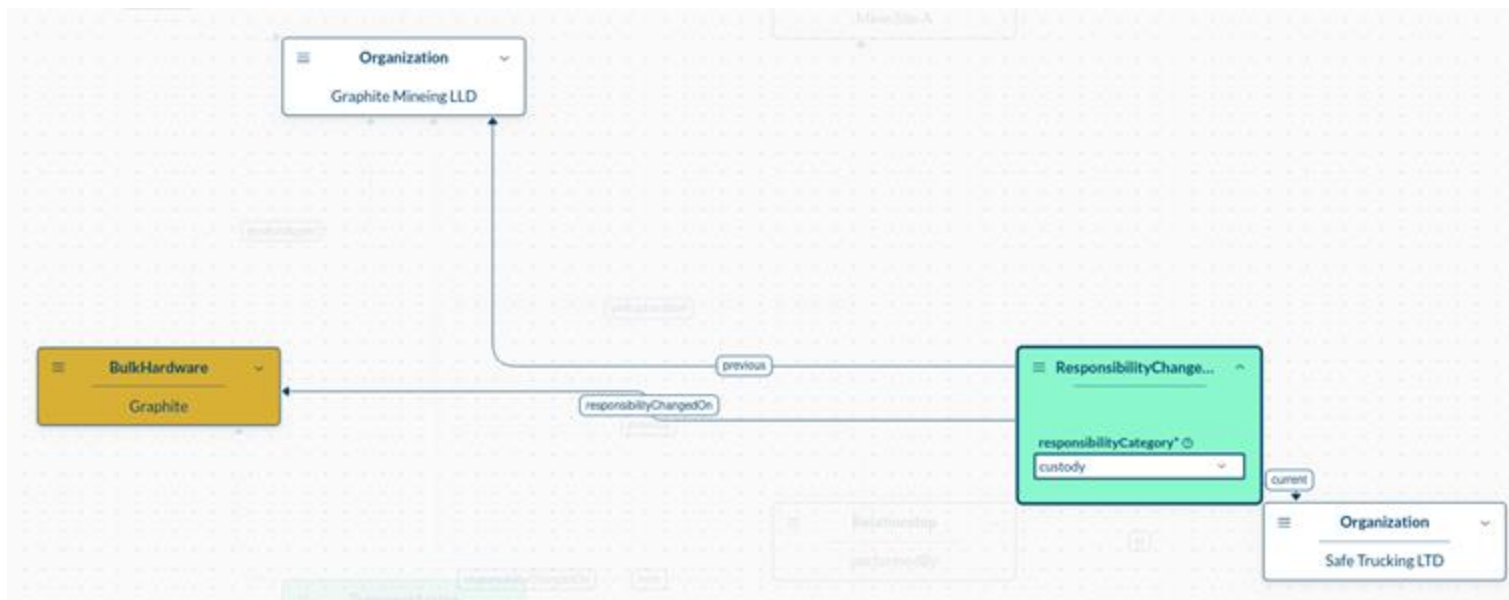


Supply Chain And SPDX

Example Supply Chain



Showing a Custody of a
assist/product change



Supply Chain and SPDX

Example Supply Chain



Full graph



Parallel Regularly Needs

Cybersecurity Resilience Act

- Widespread Vulnerabilities
- Fragmented Regulations
- Global, Cross-Border Risks
- Economic and National Security
- Consumer Protection and Confidence
- Clear, Uniform Standards

In summary, the Cyber Resilience Act aims to ensure that all digital products in the EU are secure by design and stay secure throughout their lifecycle to reduce vulnerabilities, unifying standards, and enhancing resilience against ever-evolving cyber threats.

Digital Product Passport

- Transparency and Traceability
- Sustainability and Circular Economy
- Consumer Empowerment
- Regulatory Compliance
- Business Value: Commitment to transparency and sustainability.
- Innovation and Collaboration

In summary, the Digital Product Passport aims to bridge the gap between growing demands for transparency, regulatory requirements, and the need for more sustainable.

DPP Requirements Summary

Supply Chain Element Mapping to DPP Requirements



Graphite Digital Product Passport Template

1. Mine and Extraction Details

- Mine Name:
- Mine Location (GPS):
- Ownership/Operator:
- Extraction Method:
- License/Permit Number:
- Date of Extraction:
- Environmental & Social Compliance:

2. Raw Material Data

- Material Type: Natural Graphite
- Grade/Purity:
- Batch Number:
- Volume Extracted (tonnes):

3. Transportation

- Transport Method: (e.g., Truck, Rail)
- Route Description:
- Shipping Company:
- CO₂ Emissions (estimate):
- Date/Shipped To:

4. Processing/Refining

- Processing Facility Name:
- Facility Location:
- Processing Method:
- Chemicals/Inputs Used:
- Energy Use:
- Waste/Byproducts:
- Certifications:

5. Manufacturing/Assembly

- Manufacturer Name:
- Manufacturing Facility Location:
- Product Type: (e.g., Battery Anode)
- Serial/Batch Number:
- Date of Manufacture:

6. Distribution & End Use

- Shipment Details:
- Recipient Company Name:
- Intended Product Use: (e.g., EV Battery)
- Product Passport Linkage:

7. Sustainability Metrics

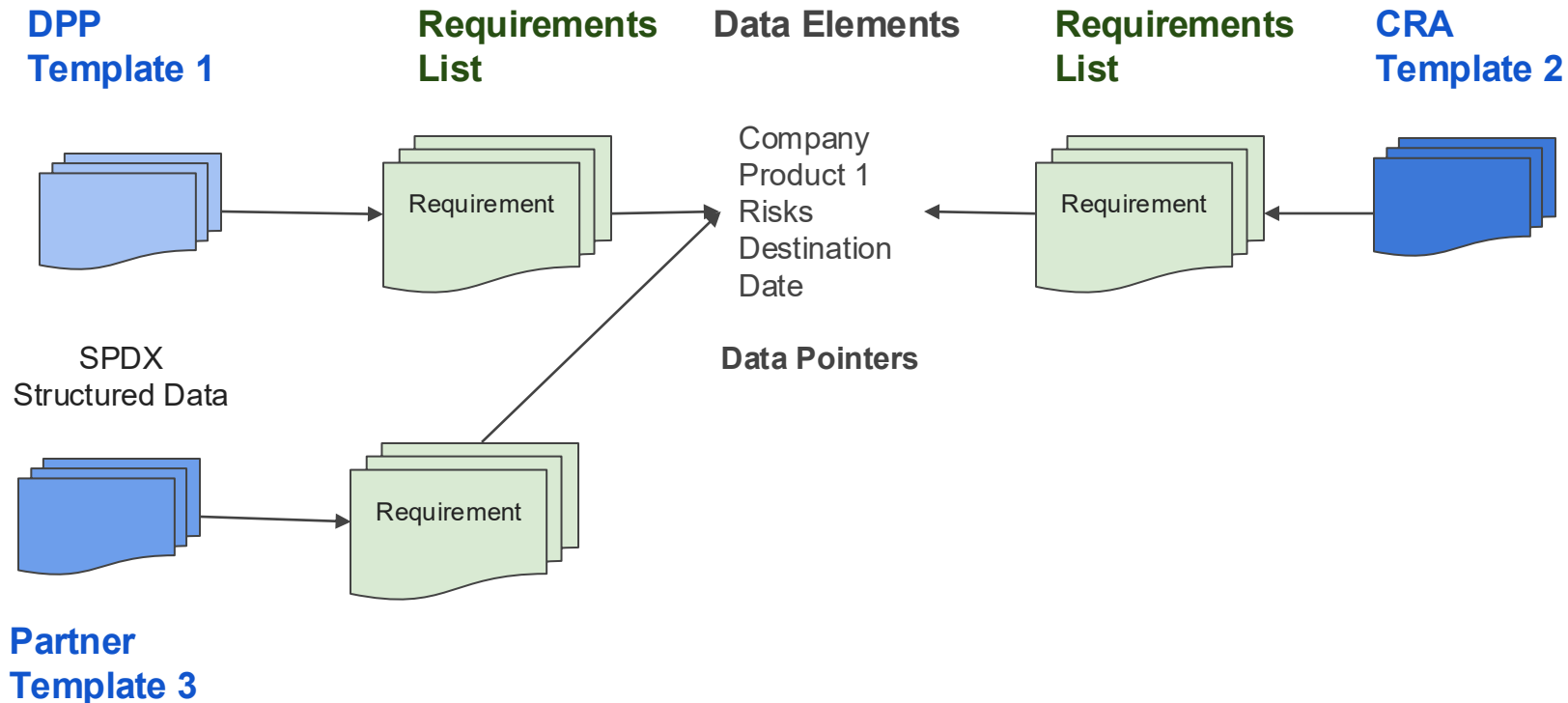
- Total Carbon Footprint:
- Water Use:
- Recyclability Information:
- End-of-Life Instructions:

8. Compliance & Traceability

- Regulatory Documents:
- ESG Audit Certificate:
- Chain of Custody Record:
- Digital Product Passport ID (QR/NFC):

Requirements Templates

One Set of Data to Multiple Requirements

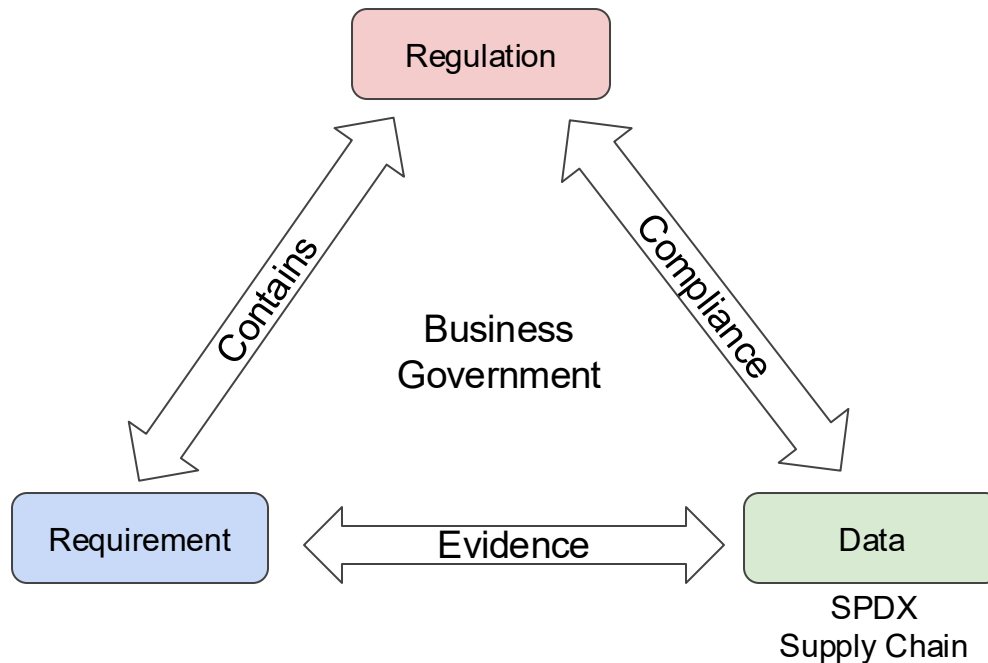


Partner
Template 3



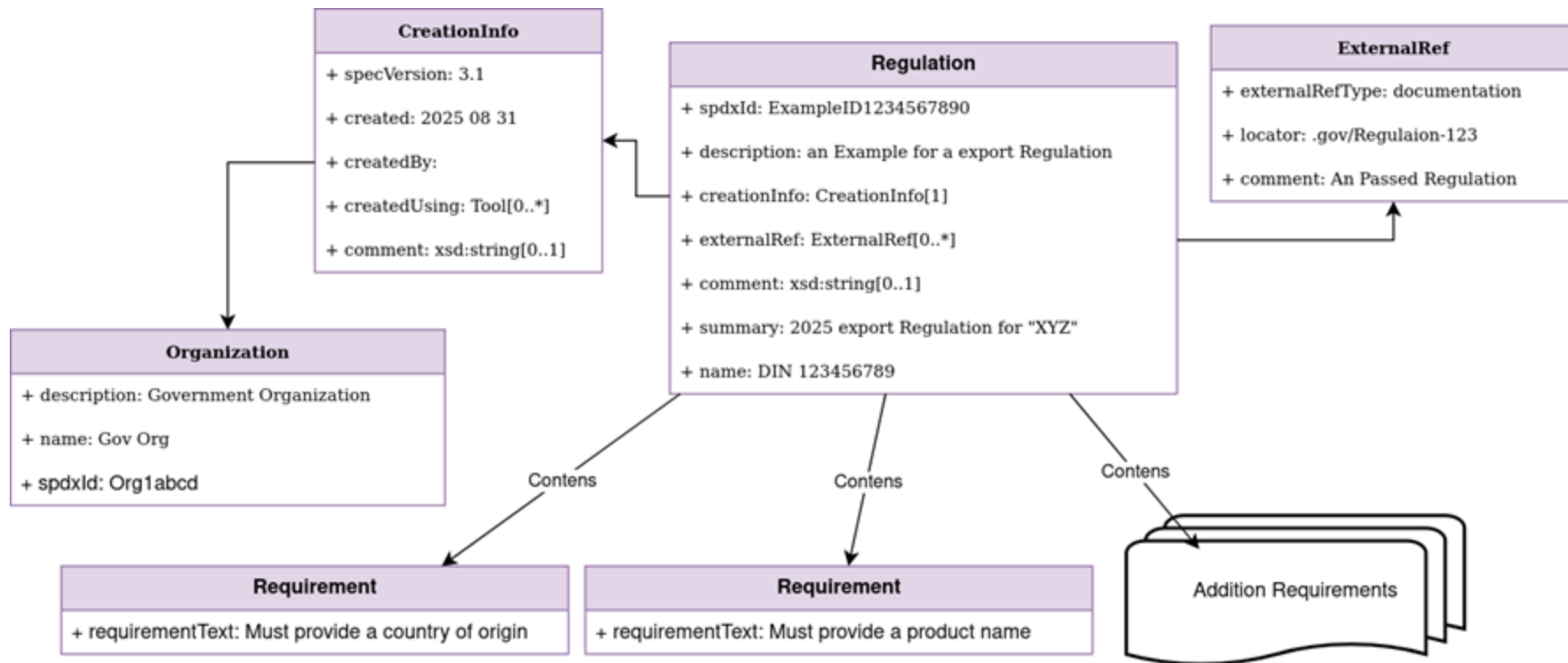
Regulatory Chain

Supporting Distributed Responsibilities



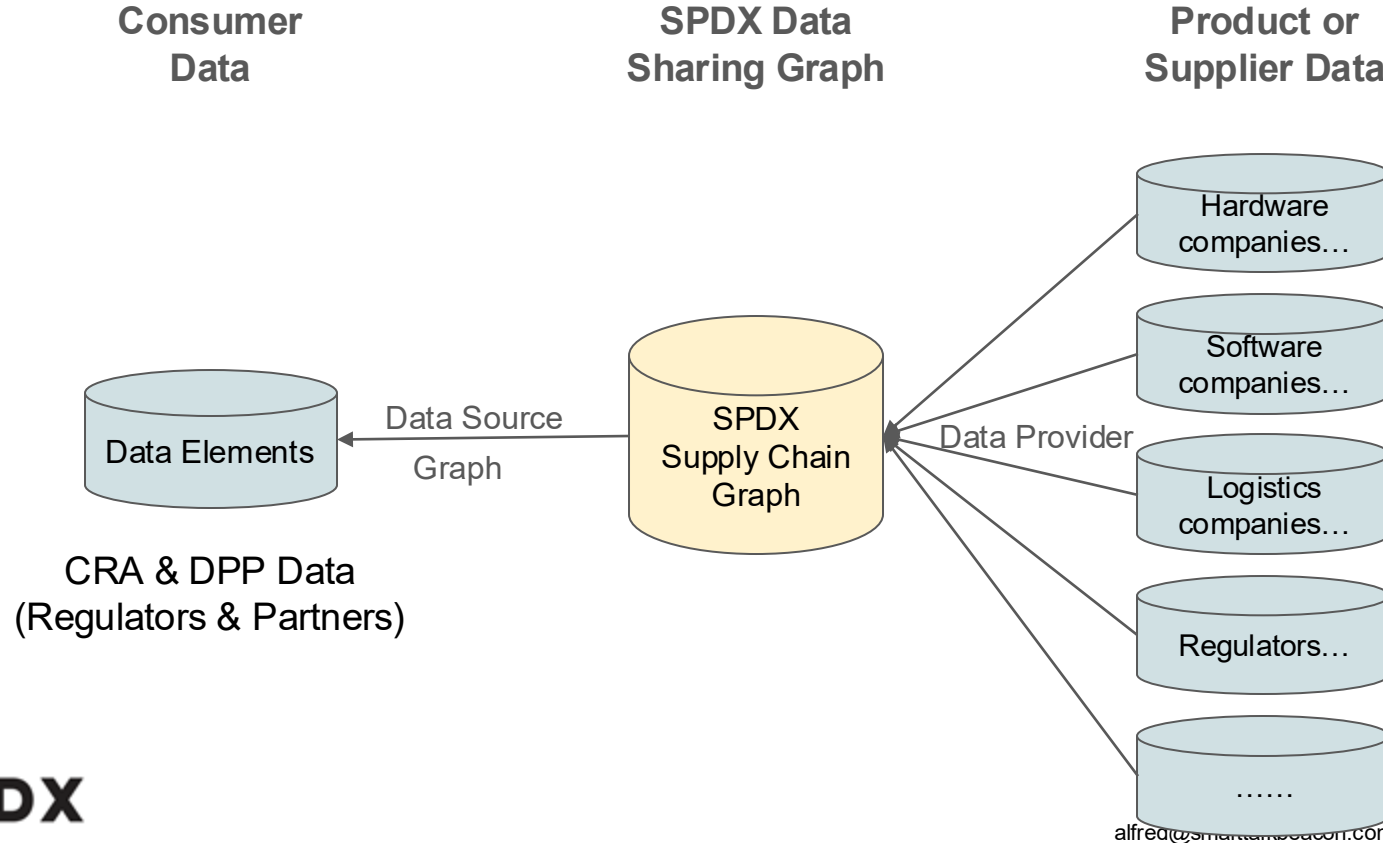
Assembling a Regulatory Requirement

SPDX Elements Used to Define Regulation



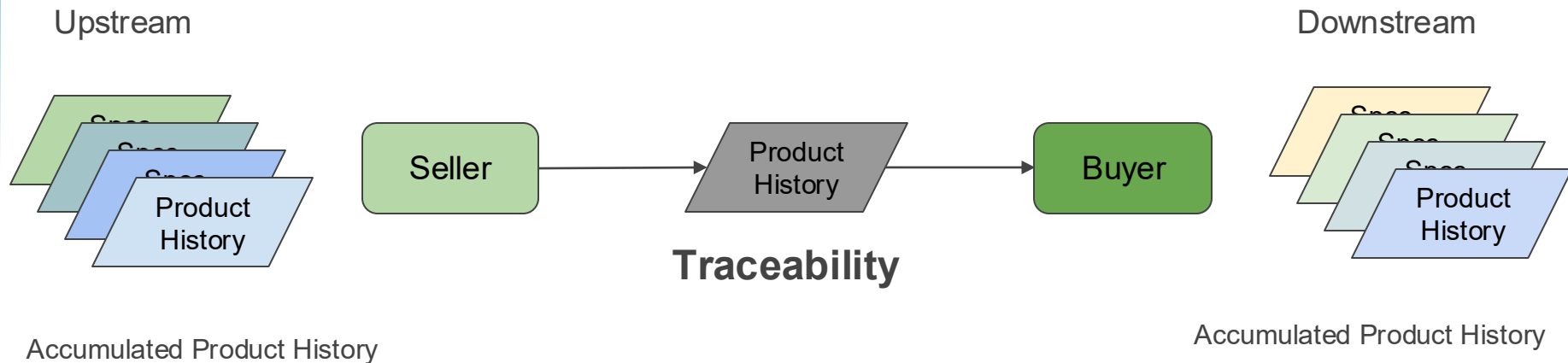
SPDX makes it easy to read and link

Use Case: Augmenting Information



Product Traceability

Ownership and Custody Tracking - Origin to End of Life



Uranium Mining Metadata



Disposal Site

Structured Document definition (One example)

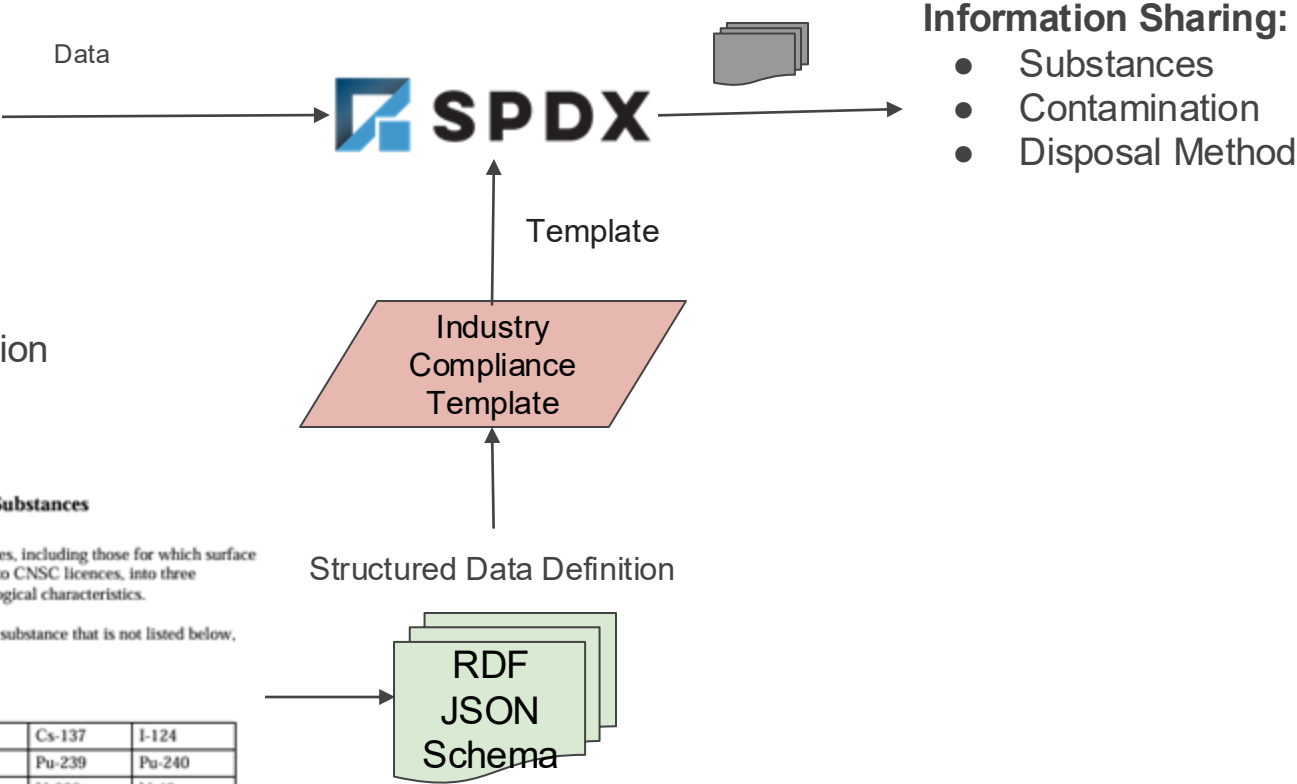
Appendix Y: Classes of Nuclear Substances

The following table organizes a number of common nuclear substances, including those for which surface contamination and waste disposal limits are typically incorporated into CNSC licences, into three classes -Class A, Class B, or Class C- on the basis of common radiological characteristics.

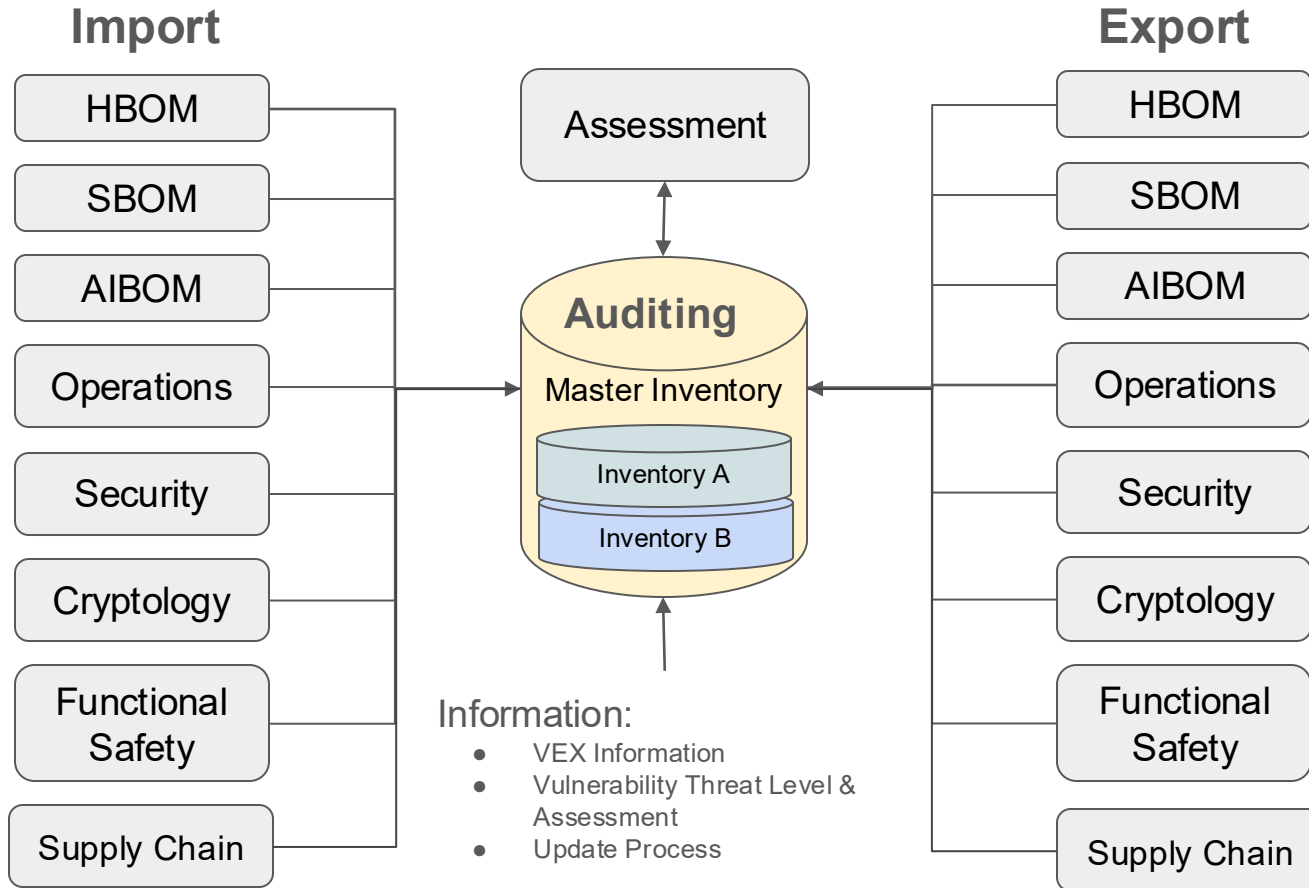
To find out the classification, for regulatory purposes, of any nuclear substance that is not listed below, contact a CNSC Licensing Specialist at 1-888-229-2672.

CLASS A: 3 Bq/cm² controlled, 0.3 Bq/cm² public

Ag-110m	Bi-210	Co-56	Co-60	Cs-134	Cs-137	I-124
Lu-177m	Mn-52	Na-22	Po-210	Pu-238	Pu-239	Pu-240
Sb-124	Sc-46	Sr-82	U-234	U-235	U-238	V-48
Zn-65	All alpha emitters and their daughter isotopes					



SPDX Import and Export





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SPDX

Hardware